CSE 579
Module 6 Graded Assignment
Template for clingo Work

Problem 1

Input Program	Hint: you only need one program with a new term, whose value will be assigned to 4 or 5 in the command line. %%%%%%%%%%%%%%%%%% % File: pa2_problem1.lp: Blocks World with Table Size Constraint %%%%%%%%%%%%%%%%%% #const m=5. % maxstep - will vary in testing #const grippers=2. % number of grippers (use 2 as per hint) #const n=4. % maximum blocks on table at once %%%%%%%%%%%%%%%%%%%%%%%%%%%% % sort and object declaration %%%%%%%%%%%%%%%%%%%%%%%%%%%% % every block is a location location(B) :- block(B). % the table is a location location(table). %%%%%%%%%%%%%%%%%%%%%%%%%% % state description %%%%%%%%%%%%%%%%%%%%%%%%%% % two blocks can't be on the same block at the same time :- 2{on(BB,B,T)}, block(B), T = 0..m. % NEW CONSTRAINT FOR PROBLEM 1: no more than n blocks on table at once :- n+1{on(B,table,T)}, T = 0..m. %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% % effect and preconditions of action %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% % effect of moving a block on(B,L,T+1) :- move(B,L,T). % concurrent actions are limited by num of grippers :- not {move(BB,LL,T)} grippers, T = 0..m-1. % a block can be moved only when it is clear

	:- move(B,L,T), on(B1,B,T). % a block can't be moved onto a block that is being moved also :- move(B,B1,T), move(B1,L,T). %%%%%%%%%%%%%%%%%%%%%%%%%% % domain independent axioms %%%%%%%%%%%%%%%%%%%%%%%%%% % fluents are initially exogenous 1{on(B,LL,0):location(LL)}1 :- block(B). % uniqueness and existence of value constraints :- not 1{on(B,LL,T)}1, block(B), T=1..m. % actions are exogenous {move(B,L,T)} :- block(B), location(L), T = 0..m-1. % commonsense law of inertia {on(B,L,T+1)} :- on(B,L,T), T < m. #show move/3. %%%%%%%%%%%%%%%%%% % File: pa2_problem1_scenario.lp %%%%%%%%%%%%%%%%%% block(1..6). % initial state :- not on(1,2,0; 2,table,0; 3,4,0; 4,table,0; 5,6,0; 6,table,0). % goal :- not on(3,2,m; 2,1,m; 1,table,m; 6,5,m; 5,4,m; 4,table,m).
Command Lines	You should write multiple command lines below. clingo pa2_problem1.lp pa2_problem1_scenario.lp -c m=4 -c grippers=2 -c n=3 clingo pa2_problem1.lp pa2_problem1_scenario.lp -c m=5 -c grippers=2 -c n=3 clingo pa2_problem1.lp pa2_problem1_scenario.lp -c m=3 -c grippers=2 -c n=4 clingo pa2_problem1.lp pa2_problem1_scenario.lp -c m=4 -c grippers=2 -c n=4

Outputs of clingo	You should write multiple outputs, one for each command. These outputs serve as the evidences of your answer to the following question. Hint 1: Let n be the maximal number of blocks that can be placed directly on the table. There should be 2 command lines and outputs for n=4, where • the 1st command line and output show k steps are not enough and • the 2nd command line and output show k+1 steps are enough. Similarly, there should be another 2 command lines and outputs for n=5. Hint 2: Use only 2 grippers. (clingo-env) C:\Users\casey\clingo> clingo pa2_problem1.lp pa2_problem1_scenario.lp -c m=4 -c grippers=2 -c n=3 clingo version 5.8.0 Reading from pa2_problem1.lp ... Solving... UNSATISFIABLE Models : 0 Calls : 1 Time : 0.008s (Solving: 0.00s 1st Model: 0.00s Unsat: 0.00s) CPU Time : 0.000s (clingo-env) C:\Users\casey\clingo> clingo pa2_problem1.lp pa2_problem1_scenario.lp -c m=5 -c grippers=2 -c n=3 clingo version 5.8.0 Reading from pa2_problem1.lp ... Solving... Answer: 1 (Time: 0.009s) move(3,6,0) move(5,4,0) move(1,5,1) move(1,table,2) move(2,5,2) move(2,1,3) move(3,5,3) move(3,2,4) move(6,5,4) SATISFIABLE Models : 1+ Calls : 1 Time : 0.011s (Solving: 0.00s 1st Model: 0.00s Unsat: 0.00s) CPU Time : 0.016s (clingo-env) C:\Users\casey\clingo> clingo pa2_problem1.lp pa2_problem1_scenario.lp -c m=3 -c grippers=2 -c n=4 clingo version 5.8.0 Reading from pa2_problem1.lp ... Solving...

<table>
<tr><td></td><td>UNSATISFIABLE

Models : 0
Calls : 1
Time : 0.006s (Solving: 0.00s 1st Model: 0.00s Unsat: 0.00s)
CPU Time : 0.016s

(clingo-env) C:\Users\casey\clingo> clingo pa2_problem1.lp pa2_problem1_scenario.lp -c m=4 -c grippers=2 -c n=4
clingo version 5.8.0
Reading from pa2_problem1.lp ...
Solving...
Answer: 1 (Time: 0.008s)
move(1,table,0) move(2,4,1) move(3,table,1) move(2,1,2) move(5,4,2) move(3,2,3) move(6,5,3)
SATISFIABLE

Models : 1+
Calls : 1
Time : 0.009s (Solving: 0.00s 1st Model: 0.00s Unsat: 0.00s)
CPU Time : 0.000s</td></tr>
<tr><td>Answer
to Questions</td><td>Fill in the following table that lists the minimum number of steps to solve the modified block world problem for different values of n, where n is the maximal number of blocks that can be placed directly on the table.

<table><tr><th>n</th><th>Number of steps</th><th>Number of Moves</th></tr><tr><td>3</td><td>5</td><td>9</td></tr><tr><td>4</td><td>4</td><td>7</td></tr></table></td></tr>
</table>

Problem 2

Input Program	Hint 1: You don't need to represent any scenario since you want to find out all possible valid states with 6 blocks. Think about the value of m. Hint 2: You don't need to consider the limitation of the maximum number of blocks on the table. That's only required in Problem 1. %%%%%%%%%%%%%%%%%% % File: pa2_problem2.lp: Blocks World - Valid States Only %%%%%%%%%%%%%%%%%% #const m=0. %%%%%%%%%%%%%%%%%%%%%%%%%%%% % sort and object declaration %%%%%%%%%%%%%%%%%%%%%%%%%%%% % every block is a location location(B) :- block(B). % the table is a location location(table). %%%%%%%%%%%%%%%%%%%%%%%%%% % state description %%%%%%%%%%%%%%%%%%%%%%%%%% % two blocks can't be on the same block at the same time :- 2{on(BB,B,T)}, block(B), T = 0..m. %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% % NEW FOR PROBLEM 2: Define "above" relationship %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% % Block A is above block B if A is directly on B above(A,B,T) :- on(A,B,T), block(B). % Block A is above block B if A is above some block C, and C is above B (transitive) above(A,B,T) :- above(A,C,T), above(C,B,T), block(A), block(B), block(C). % Constraint: Nothing can be above itself (prevents cycles and self-references) :- above(B,B,T), block(B), T = 0..m. %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% % effect and preconditions of action %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% % effect of moving a block

	on(B,L,T+1) :- move(B,L,T). % concurrent actions are limited by num of grippers :- not {move(BB,LL,T)} grippers, T = 0..m-1. % a block can be moved only when it is clear :- move(B,L,T), on(B1,B,T). % a block can't be moved onto a block that is being moved also :- move(B,B1,T), move(B1,L,T). %%%%%%%%%%%%%%%%%%%%%%%%%% % domain independent axioms %%%%%%%%%%%%%%%%%%%%%%%%%% % fluents are initially exogenous 1{on(B,LL,0):location(LL)}1 :- block(B). % uniqueness and existence of value constraints :- not 1{on(B,LL,T)}1, block(B), T=1..m. % actions are exogenous {move(B,L,T)} :- block(B), location(L), T = 0..m-1. % commonsense law of inertia {on(B,L,T+1)} :- on(B,L,T), T < m. #show on/3. %%%%%%%%%%%%%%%%%% % File: pa2_problem2_scenario.lp %%%%%%%%%%%%%%%%%% block(1..6).
Command Line	clingo pa2_problem2.lp pa2_problem2_scenario.lp 0
Output of clingo	(clingo-env) C:\Users\casey\clingo> clingo pa2_problem2.lp pa2_problem2_scenario.lp 0 clingo version 5.8.0 Reading from pa2_problem2.lp ... Solving... on(1,6,0) on(2,5,0) on(3,4,0) on(4,1,0) on(6,2,0) on(5,table,0) Answer: 4051 (Time: 22.572s) on(1,4,0) on(2,5,0) on(3,1,0) on(4,6,0) on(6,2,0) on(5,table,0) SATISFIABLE Models : 4051 Calls : 1 Time : 22.573s (Solving: 22.57s 1st Model: 0.00s Unsat: 0.00s) CPU Time : 0.781s

Answer to Questions	How many valid states are there when there are 6 blocks? (Note that the limitation of blocks introduced in question 1 is not considered here.) 4051 states

Problem 3

Reading: A plan may allow multiple actions happening at the same time, e.g., when we have multiple robots working together to increase efficiency. However, if there is a little bit delay on one action, then we may get unexpected results. For example, when 2 robots are moving 2 adjacent blocks to the left at the same time, if there is a delay for the robot on the left-hand side, then these 2 robots may hit with each other. To make sure that our plan will get the expected result, we introduce the restriction "serializable" on the actions happening at the same time. This restriction simply says that, even if some actions in the same time stamp happen in serial with arbitrary ordering, the final result would be the same.

Input Program	Hint: the number of grippers is unlimited, meaning that you can have as many movements as you want as far as the movements are serializable. %%%%%%%%%%%%%%%%%% % File: pa2_problem3.lp: Blocks World - Serializable Plans Only %%%%%%%%%%%%%%%%%% #const m=8. %%%%%%%%%%%%%%%%%%%%%%%%%%%% % sort and object declaration %%%%%%%%%%%%%%%%%%%%%%%%%%%% % every block is a location location(B) :- block(B). % the table is a location location(table). %%%%%%%%%%%%%%%%%%%%%%%%%% % state description %%%%%%%%%%%%%%%%%%%%%%%%%% % two blocks can't be on the same block at the same time :- 2{on(BB,B,T)}, block(B), T = 0..m. %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% % effect and preconditions of action %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% % effect of moving a block on(B,L,T+1) :- move(B,L,T). % a block can be moved only when it is clear :- move(B,L,T), on(B1,B,T). % a block can't be moved onto a block that is being moved also

	:- move(B,B1,T), move(B1,L,T). %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% % NEW FOR PROBLEM 3: Serializable constraint %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% % Cannot move A onto block B if something C is already on B at time T :- move(A,B,T), on(C,B,T), block(B), A != C. %%%%%%%%%%%%%%%%%%%%%%%%%%%% % domain independent axioms %%%%%%%%%%%%%%%%%%%%%%%%%%%% % fluents are initially exogenous 1{on(B,LL,0):location(LL)}1 :- block(B). % uniqueness and existence of value constraints :- not 1{on(B,LL,T)}1, block(B), T=1..m. % actions are exogenous {move(B,L,T)} :- block(B), location(L), T = 0..m-1. % commonsense law of inertia {on(B,L,T+1)} :- on(B,L,T), T < m. #show move/3. %%%%%%%%%%%%%%%%%% % File: pa2_problem3_scenario.lp %%%%%%%%%%%%%%%%%% block(a;b;c;d;e;f;g;h;i;j;k;l;m1;n;o). % initial state :- not on(m1,table,0; l,m1,0; a,l,0; b,a,0; c,b,0; o,table,0; n,o,0; d,n,0; e,d,0; j,e,0; k,j,0; f,table,0; g,f,0; h,g,0; i,h,0). % goal :- not on(e,j,m; a,e,m; n,a,m; i,d,m; h,i,m; m1,h,m; o,m1,m; k,g,m; c,k,m; b,c,m; l,b,m).
Command Line	Please only show the command line that outputs the minimal length plan. clingo pa2_problem3.lp pa2_problem3_scenario.lp -c m=8
Output of clingo	(clingo-env) C:\Users\casey\clingo> clingo pa2_problem3.lp pa2_problem3_scenario.lp -c m=8 clingo version 5.8.0 Reading from pa2_problem3.lp ...

	Solving... Answer: 1 (Time: 0.095s) move(c,table,0) move(i,table,0) move(k,table,0) move(b,table,1) move(h,table,1) move(j,table,1) move(a,table,2) move(e,table,2) move(g,table,2) move(d,table,3) move(k,g,3) move(l,table,3) move(c,k,4) move(e,j,4) move(i,d,4) move(n,table,4) move(b,c,5) move(h,i,5) move(a,e,6) move(l,b,6) move(m1,h,6) move(n,a,7) move(o,m1,7) SATISFIABLE Models : 1+ Calls : 1 Time : 0.111s (Solving: 0.02s 1st Model: 0.00s Unsat: 0.00s) CPU Time : 0.094s

Problem 4

Input Program	%%%%%%%%%%%%%%%%%% % File: pa2_problem4.lp: Blocks World - Optimal Plans (Minimize Actions) %%%%%%%%%%%%%%%%%% #const m=8. #const grippers=10. %%%%%%%%%%%%%%%%%%%%%%%%%%%% % sort and object declaration %%%%%%%%%%%%%%%%%%%%%%%%%%%% % every block is a location location(B) :- block(B). % the table is a location location(table). %%%%%%%%%%%%%%%%%%%%%%%%%% % state description %%%%%%%%%%%%%%%%%%%%%%%%%% % two blocks can't be on the same block at the same time :- 2{on(BB,B,T)}, block(B), T = 0..m. %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% % effect and preconditions of action %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% % effect of moving a block on(B,L,T+1) :- move(B,L,T). % concurrent actions are limited by num of grippers :- not {move(BB,LL,T)} grippers, T = 0..m-1. % a block can be moved only when it is clear :- move(B,L,T), on(B1,B,T). % a block can't be moved onto a block that is being moved also :- move(B,B1,T), move(B1,L,T). %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% % Serializable constraint (from Problem 3) %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% % Cannot move A onto block B if something C is already on B at time T :- move(A,B,T), on(C,B,T), block(B), A != C. %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% % NEW FOR PROBLEM 4: Minimize total number of actions

	%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% #minimize { 1,B,L,T : move(B,L,T) }. %%%%%%%%%%%%%%%%%%%%%%%%%%%% % domain independent axioms %%%%%%%%%%%%%%%%%%%%%%%%%%%% % fluents are initially exogenous 1{on(B,LL,0):location(LL)}1 :- block(B). % uniqueness and existence of value constraints :- not 1{on(B,LL,T)}1, block(B), T=1..m. % actions are exogenous {move(B,L,T)} :- block(B), location(L), T = 0..m-1. % commonsense law of inertia {on(B,L,T+1)} :- on(B,L,T), T < m. #show move/3. %%%%%%%%%%%%%%%%%% % File: pa2_problem4_scenario.lp %%%%%%%%%%%%%%%%%% block(a;b;c;d;e;f;g;h;i;j;k;l;m1;n;o). % initial state :- not on(m1,table,0; l,m1,0; a,l,0; b,a,0; c,b,0; o,table,0; n,o,0; d,n,0; e,d,0; j,e,0; k,j,0; f,table,0; g,f,0; h,g,0; i,h,0). % goal :- not on(e,j,m; a,e,m; n,a,m; i,d,m; h,i,m; m1,h,m; o,m1,m; k,g,m; c,k,m; b,c,m; l,b,m).
Command Line	You should write multiple command lines below. clingo pa2_problem4.lp pa2_problem4_scenario.lp -c m=8 -c grippers=10 -t 8 clingo pa2_problem4.lp pa2_problem4_scenario.lp -c m=9 -c grippers=10 -t 8 clingo pa2_problem4.lp pa2_problem4_scenario.lp -c m=10 -c grippers=10 -t 8 clingo pa2_problem4.lp pa2_problem4_scenario.lp -c m=11 -c grippers=10 -t 8

Output of clingo	You should write multiple outputs, one for each command. These outputs serve as the evidences of your answer to the question below. (clingo-env) C:\Users\casey\clingo> clingo pa2_problem4.lp pa2_problem4_scenario.lp -c m=8 -c grippers=10 -t 8 clingo version 5.8.0 Reading from pa2_problem4.lp ... Solving... Progression : [1;inf] (Time: 0.109s) Progression : [2;inf] (Time: 0.109s) Progression : [3;inf] (Time: 0.110s) Answer: 1 (Time: 0.110s) move(c,table,0) move(i,table,0) move(k,table,0) move(b,table,1) move(h,table,1) move(j,table,1) move(a,table,2) move(e,table,2) move(g,table,2) move(d,table,3) move(e,j,3) move(k,g,3) move(l,table,3) move(i,d,4) move(n,table,4) move(c,k,5) move(h,i,5) move(a,e,6) move(b,c,6) move(m1,h,6) move(l,b,7) move(n,a,7) move(o,m1,7) Optimization: 23 Progression : [4;23] (Error: 4.75 Time: 0.115s) Progression : [5;23] (Error: 3.6 Time: 0.116s) Answer: 2 (Time: 0.116s) move(c,i,0) move(k,table,0) move(b,table,1) move(c,table,1) move(j,table,1) move(a,b,2) move(e,j,2) move(i,table,2) move(d,table,3) move(h,c,3) move(l,table,3) move(a,e,4) move(h,l,4) move(i,d,4) move(k,g,4) move(c,k,5) move(h,i,5) move(b,c,6) move(m1,h,6) move(n,a,6) move(l,b,7) move(o,m1,7) Optimization: 22 Progression : [6;22] (Error: 2.66667 Time: 0.120s) Progression : [7;22] (Error: 2.14286 Time: 0.120s) Progression : [8;22] (Error: 1.75 Time: 0.121s) Progression : [9;22] (Error: 1.44444 Time: 0.121s) Progression : [10;22] (Error: 1.2 Time: 0.122s) Progression : [11;22] (Error: 1 Time: 0.122s) Progression : [12;22] (Error: 0.833333 Time: 0.122s) Answer: 3 (Time: 0.123s) move(k,i,0) move(c,table,1) move(j,table,1) move(k,table,1) move(b,table,2) move(e,k,2) move(i,c,2) move(d,table,3) move(e,j,3) move(h,b,3) move(a,e,4) move(i,d,4) move(k,g,4) move(c,k,5) move(h,i,5) move(l,table,5) move(n,a,5) move(b,c,6) move(m1,h,6) move(l,b,7) move(o,m1,7) Optimization: 21 Answer: 4 (Time: 0.127s) move(i,table,0) move(k,table,0) move(c,table,1) move(j,table,1) move(b,table,2) move(e,j,2) move(h,table,2) move(d,table,3) move(k,g,3) move(a,e,4)

	move(c,k,4) move(i,d,4) move(b,c,5) move(h,i,5) move(l,table,5) move(l,b,6) move(m1,h,6) move(n,a,6) move(n,a,7) move(o,m1,7) Optimization: 20 Progression : [13;20] (Error: 0.538462 Time: 0.132s) Progression : [14;20] (Error: 0.428571 Time: 0.132s) Progression : [15;20] (Error: 0.333333 Time: 0.133s) Progression : [16;20] (Error: 0.25 Time: 0.133s) Progression : [17;20] (Error: 0.176471 Time: 0.134s) Progression : [18;20] (Error: 0.111111 Time: 0.135s) Answer: 5 (Time: 0.137s) move(c,table,0) move(i,table,0) move(k,table,0) move(h,table,1) move(j,table,1) move(b,table,2) move(e,j,2) move(k,g,2) move(a,e,3) move(c,k,3) move(d,table,3) move(b,c,4) move(i,d,4) move(h,i,5) move(l,b,5) move(n,a,5) move(m1,h,6) move(o,m1,7) Optimization: 18 OPTIMUM FOUND Models : 5 Optimum : yes Optimization : 18 Calls : 1 Time : 0.140s (Solving: 0.04s 1st Model: 0.01s Unsat: 0.00s) CPU Time : 0.188s Threads : 8 (Winner: 1) (clingo-env) C:\Users\casey\clingo> clingo pa2_problem4.lp pa2_problem4_scenario.lp -c m=9 -c grippers=10 -t 8 clingo version 5.8.0 Reading from pa2_problem4.lp ... Solving... Progression : [1;inf] (Time: 0.111s) Answer: 1 (Time: 0.111s) move(c,table,0) move(i,table,0) move(k,table,0) move(b,table,1) move(h,table,1) move(j,table,1) move(a,table,2) move(e,table,2) move(g,table,2) move(d,table,3) move(k,g,3) move(l,table,3) move(c,k,4) move(i,d,4) move(n,table,4) move(e,j,5) move(h,i,5) move(a,e,6) move(b,c,6) move(m1,h,6) move(l,b,7) move(n,a,7) move(o,m1,7) Optimization: 23 Progression : [2;23] (Error: 10.5 Time: 0.115s) Progression : [3;23] (Error: 6.66667 Time: 0.116s) Progression : [4;23] (Error: 4.75 Time: 0.116s) Progression : [5;23] (Error: 3.6 Time: 0.117s) Progression : [6;23] (Error: 2.83333 Time: 0.117s)

	Progression : [7;23] (Error: 2.28571 Time: 0.118s) Progression : [8;23] (Error: 1.875 Time: 0.118s) Progression : [9;23] (Error: 1.55556 Time: 0.119s) Progression : [10;23] (Error: 1.3 Time: 0.119s) Answer: 2 (Time: 0.120s) move(k,c,0) move(i,table,1) move(j,table,1) move(k,table,1) move(c,i,2) move(h,table,2) move(b,table,3) move(e,j,3) move(k,g,3) move(a,e,4) move(c,k,4) move(d,table,4) move(b,c,5) move(i,d,5) move(n,a,5) move(h,i,6) move(l,b,6) move(n,o,6) move(m1,h,7) move(n,table,7) move(n,a,8) move(o,m1,8) Optimization: 22 Answer: 3 (Time: 0.125s) move(i,table,0) move(c,i,1) move(h,table,1) move(k,table,1) move(b,h,2) move(c,table,2) move(j,table,2) move(k,g,2) move(b,i,3) move(c,k,3) move(e,j,3) move(a,e,4) move(b,c,4) move(d,table,4) move(i,d,5) move(l,b,5) move(n,table,5) move(h,i,6) move(m1,h,7) move(n,a,7) move(o,m1,8) Optimization: 21 Progression : [11;21] (Error: 0.909091 Time: 0.129s) Progression : [12;21] (Error: 0.75 Time: 0.129s) Progression : [13;21] (Error: 0.615385 Time: 0.130s) Progression : [14;21] (Error: 0.5 Time: 0.130s) Progression : [15;21] (Error: 0.4 Time: 0.132s) Answer: 4 (Time: 0.133s) move(i,c,0) move(h,table,1) move(i,table,1) move(k,table,1) move(c,h,2) move(j,table,2) move(k,g,2) move(b,i,3) move(c,k,3) move(e,j,3) move(a,e,4) move(b,c,4) move(d,table,4) move(h,n,5) move(i,d,5) move(l,b,5) move(h,i,6) move(m1,h,7) move(n,a,7) move(o,m1,8) Optimization: 20 Progression : [16;20] (Error: 0.25 Time: 0.136s) Answer: 5 (Time: 0.141s) move(i,table,0) move(k,table,0) move(h,table,1) move(j,table,1) move(e,j,2) move(k,g,2) move(c,k,3) move(d,table,3) move(b,c,4) move(i,d,4) move(a,e,5) move(h,i,5) move(l,b,6) move(n,a,6) move(m1,h,7) move(o,m1,8) Optimization: 16 OPTIMUM FOUND Models : 5 Optimum : yes Optimization : 16 Calls : 1 Time : 0.144s (Solving: 0.04s 1st Model: 0.00s Unsat: 0.00s)

	CPU Time : 0.313s Threads : 8 (Winner: 1) (clingo-env) C:\Users\casey\clingo> clingo pa2_problem4.lp pa2_problem4_scenario.lp -c m=10 -c grippers=10 -t 8 clingo version 5.8.0 Reading from pa2_problem4.lp ... Solving... Progression : [1;inf] (Time: 0.118s) Answer: 1 (Time: 0.119s) move(c,table,0) move(i,table,0) move(k,table,0) move(b,table,1) move(h,table,1) move(j,table,1) move(a,table,2) move(e,table,2) move(g,table,2) move(d,table,3) move(l,table,3) move(n,table,4) move(i,d,5) move(k,g,5) move(c,k,6) move(e,j,6) move(h,i,6) move(a,e,7) move(b,c,7) move(m1,h,7) move(l,b,8) move(n,a,8) move(o,m1,8) Optimization: 23 Progression : [2;23] (Error: 10.5 Time: 0.123s) Progression : [3;23] (Error: 6.66667 Time: 0.123s) Progression : [4;23] (Error: 4.75 Time: 0.124s) Progression : [5;23] (Error: 3.6 Time: 0.124s) Progression : [6;23] (Error: 2.83333 Time: 0.125s) Progression : [7;23] (Error: 2.28571 Time: 0.125s) Progression : [8;23] (Error: 1.875 Time: 0.126s) Progression : [9;23] (Error: 1.55556 Time: 0.128s) Progression : [10;23] (Error: 1.3 Time: 0.129s) Progression : [11;23] (Error: 1.09091 Time: 0.130s) Progression : [12;23] (Error: 0.916667 Time: 0.132s) Answer: 2 (Time: 0.134s) move(c,i,0) move(k,table,0) move(b,table,1) move(c,table,1) move(j,table,1) move(a,table,2) move(e,j,2) move(i,k,2) move(l,table,3) move(a,m1,4) move(d,table,4) move(h,l,4) move(i,d,5) move(a,e,6) move(k,g,6) move(c,k,7) move(h,i,7) move(n,a,7) move(b,c,8) move(m1,h,8) move(l,b,9) move(o,m1,9) Optimization: 22 Progression : [13;22] (Error: 0.692308 Time: 0.139s) Progression : [14;22] (Error: 0.571429 Time: 0.147s) Answer: 3 (Time: 0.153s) move(i,k,0) move(h,table,1) move(i,c,1) move(i,h,2) move(k,g,2) move(c,k,3) move(i,table,3) move(j,table,3) move(b,i,4) move(e,j,4) move(a,e,5) move(b,c,5) move(d,table,5) move(i,d,6) move(l,table,6) move(n,a,6) move(h,i,7) move(l,b,7) move(m1,h,8) move(o,m1,9) Optimization: 20 Answer: 4 (Time: 0.156s)

	move(i,k,0) move(h,table,1) move(i,c,1) move(i,table,2) move(k,g,2) move(c,k,3) move(j,table,3) move(b,i,4) move(e,j,4) move(a,e,5) move(b,c,5) move(d,table,5) move(i,d,6) move(l,table,6) move(n,a,6) move(h,i,7) move(l,b,7) move(m1,h,8) move(o,m1,9) Optimization: 19 Progression : [15;19] (Error: 0.266667 Time: 0.159s) Answer: 5 (Time: 0.165s) move(i,table,0) move(h,table,1) move(k,g,2) move(c,k,3) move(j,table,3) move(e,j,4) move(b,c,5) move(d,table,5) move(a,e,6) move(i,d,6) move(h,i,7) move(l,b,7) move(m1,h,8) move(n,a,8) move(o,m1,9) Optimization: 15 OPTIMUM FOUND Models : 5 Optimum : yes Optimization : 15 Calls : 1 Time : 0.167s (Solving: 0.05s 1st Model: 0.00s Unsat: 0.00s) CPU Time : 0.406s Threads : 8 (Winner: 1) (clingo-env) C:\Users\casey\clingo> clingo pa2_problem4.lp pa2_problem4_scenario.lp -c m=11 -c grippers=10 -t 8 clingo version 5.8.0 Reading from pa2_problem4.lp ... Solving... Progression : [1;inf] (Time: 0.142s) Progression : [2;inf] (Time: 0.142s) Answer: 1 (Time: 0.142s) move(c,table,0) move(i,table,0) move(k,table,0) move(b,table,1) move(h,table,1) move(j,table,1) move(a,table,2) move(e,table,2) move(g,table,2) move(d,table,3) move(l,table,3) move(n,table,4) move(i,d,5) move(h,i,6) move(k,g,6) move(e,j,7) move(h,table,7) move(a,e,8) move(c,k,8) move(h,i,8) move(b,c,9) move(m1,h,9) move(n,a,9) move(l,b,10) move(o,m1,10) Optimization: 25 Progression : [3;25] (Error: 7.33333 Time: 0.145s) Answer: 2 (Time: 0.145s) move(c,table,0) move(i,table,0) move(k,table,0) move(b,table,1) move(h,table,1) move(j,table,1) move(a,table,2) move(e,table,2) move(g,table,2) move(d,table,3) move(l,table,3) move(n,table,4) move(i,d,6) move(k,g,6) move(e,j,7) move(h,i,7) move(a,e,8) move(c,k,8) move(m1,h,8) move(b,c,9) move(n,a,9) move(o,m1,9) move(l,b,10)

	Optimization: 23 Progression : [4;23] (Error: 4.75 Time: 0.147s) Progression : [5;23] (Error: 3.6 Time: 0.148s) Progression : [6;23] (Error: 2.83333 Time: 0.148s) Progression : [7;23] (Error: 2.28571 Time: 0.149s) Progression : [8;23] (Error: 1.875 Time: 0.149s) Answer: 3 (Time: 0.152s) move(k,table,0) move(i,c,1) move(j,table,1) move(e,table,2) move(h,table,2) move(i,table,2) move(d,table,3) move(k,g,3) move(c,k,4) move(i,d,4) move(b,c,5) move(e,j,5) move(h,i,5) move(a,e,6) move(l,b,7) move(m1,h,9) move(n,a,9) move(o,m1,10) Optimization: 18 Progression : [9;18] (Error: 1 Time: 0.155s) Progression : [10;18] (Error: 0.8 Time: 0.156s) Progression : [11;18] (Error: 0.636364 Time: 0.158s) Progression : [12;18] (Error: 0.5 Time: 0.173s) Progression : [13;18] (Error: 0.384615 Time: 0.183s) Progression : [14;18] (Error: 0.285714 Time: 0.200s) Progression : [15;18] (Error: 0.2 Time: 0.219s) Answer: 4 (Time: 0.236s) move(i,table,0) move(h,table,1) move(k,g,2) move(j,table,3) move(c,k,4) move(e,j,4) move(b,c,5) move(d,table,5) move(i,d,6) move(a,e,7) move(h,i,7) move(l,b,8) move(m1,h,9) move(n,a,9) move(o,m1,10) Optimization: 15 OPTIMUM FOUND Models : 4 Optimum : yes Optimization : 15 Calls : 1 Time : 0.240s (Solving: 0.10s 1st Model: 0.01s Unsat: 0.00s) CPU Time : 0.969s Threads : 8 (Winner: 1) (clingo-env) C:\Users\casey\clingo>

Answer to Questions	What is the least number of actions when maxstep m is 8, 9,10 and 11?

m	least number of actions
8	18
9	16
10	15
11	15